

Module 03: Perception in High School

Learning Objectives

1. Define **Perception** within the context of High School.
2. Analyze how Perception interacts with other components of the Active Inference framework.
3. Apply specific constraints of High School to the formal definition of Perception.

Introduction

This module explores **Perception**. In the **High School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Perception is a critical component of the 8-part Active Inference spine, bridging the gap between Agents and Cognition.

Key Concepts

1. Perception as a Markov Blanket Boundary

How does Perception define the boundary between the agent and the environment?

2. Generative Models of Perception

What parameters involved in Perception must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Perception drive the perception-action loop?

Applications

In High School, we see Perception manifest in: * **Specific Example 1:** When your phone's Face ID unlocks, it performs Active Inference perception: the infrared dot projector captures a 3D map of your face (sensory input), the neural engine compares it against a stored generative model of your facial geometry, and unlocking only occurs when the prediction error falls below a security threshold -- which is why it fails when you wear a full face mask that creates too much surprise relative to the stored model. * **Specific Example 2:** A speech recognition system like Siri perceives your voice through Active Inference: it generates predictions of likely phoneme sequences based on a language model (prior expectations), compares them against the actual audio waveform (sensory evidence), and selects the transcription that minimizes total prediction error -- which is why it handles common phrases easily but struggles with unusual names where its priors are weak.

Conclusion

Understanding Perception allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.